16/4/2026

Recall:

Gamma fn.

- $\Gamma(k) = \begin{cases} \int_0^\infty e^{-x} x^{k-1}\, dx & k>0 \\ 2\int_0^\infty e^{-x^2} x^{2k-1}\, dx \end{cases}$

- $\Gamma(k) = (k-1)\Gamma(k-1)$
- $\Gamma(1) = 1$
- $\Gamma(n) = (n-1)!$

if n is a positive integer

- $\Gamma(p)\Gamma(1-p) = \dfrac{\pi}{\sin \pi p}, \quad 0<p<1$

Beta function

- $\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\, dx$
- $\beta(m,n) = \beta(n,m)$
- $\beta(m,n) = \dfrac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$

3. Evaluate $\int_0^3 \frac{x^{3/2}}{\sqrt{3-x}}\,dx \int_0^1 \frac{dx}{\sqrt{1-x^{1/4}}}$

Soln: Let

$$I_1 = \int_0^3 \frac{x^{3/2}}{\sqrt{3-x}}\,dx \quad \& \quad I_2 = \int_0^1 \frac{dx}{\sqrt{1-x^{1/4}}}$$

In I_1, we have

$$3-x = 3\left(1-\frac{x}{3}\right)$$

$\therefore$ put $\frac{x}{3} = t$

$$\Rightarrow \boxed{\begin{aligned} x &= 3t \\ dx &= 3\,dt \end{aligned}}$$

x	0	3
$t = x/3$	0	1

In I_2, put

$$x^{1/4} = t$$
$$\Rightarrow x = t^4$$
$$\Rightarrow dx = 4t^3\,dt$$

x	0	1
$t = x^{1/4}$	0	1

$$\therefore I_1 \cdot I_2 = \int_0^1 \frac{(3t)^{3/2}}{\sqrt{3-3t}} \cdot 3dt \cdot \int_0^1 \frac{4t^3\,dt}{\sqrt{1-t}}$$

$$= 3^{\frac{3}{2}+1} \int_0^1 \frac{t^{3/2}}{[3(1-t)]^{1/2}}\,dt \cdot 4\int_0^1 t^3 (1-t)^{-1/2}\,dt$$

$$= 3^{\frac{3}{2}+1-\frac{1}{2}} \int_0^1 t^{3/2}(1-t)^{-1/2}\,dt \cdot 4\,\beta\left(4, \frac{1}{2}\right)$$

$$= 36\,\beta\left(\frac{5}{2}, \frac{1}{2}\right) \cdot \beta\left(4, \frac{1}{2}\right)$$

$$= 36 \cdot \frac{\Gamma\left(\frac{5}{2}\right)\cdot\Gamma(1/2)}{\Gamma(3)} \cdot \frac{\Gamma(4)\cdot\Gamma(1/2)}{\Gamma\left(\frac{9}{2}\right)}$$

$$\Rightarrow I = 36 \cdot \frac{\frac{3}{2}\cdot\cancel{\frac{1}{2}\Gamma(1/2)}\cdot\sqrt{\pi}}{2!} \cdot \frac{3!\sqrt{\pi}}{\frac{7}{2}\cdot\frac{5}{2}\cdot\frac{3}{2}\cdot\cancel{\frac{1}{2}\Gamma(1/2)}} = \frac{432\pi}{35}$$

Result: We have

$$\int_0^{\pi/2} \sin^p\theta \cos^q\theta \, d\theta = \frac{1}{2}\beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$$

$$\Rightarrow \boxed{\beta(m,n) = 2\int_0^{\pi/2} \sin^{2m-1}\theta \cos^{2n-1}\theta \, d\theta}$$

$m = \frac{p+1}{2}, \quad n = \frac{q+1}{2}$

$\Rightarrow p = 2m-1, \quad q = 2n-1$

Eg: Evaluate

$$\int_0^{\pi/2} \sin^2\theta \cos^3\theta \, d\theta = \frac{1}{2}\beta\left(\frac{3}{2}, 2\right)$$

$p = 2, \quad q = 3$

$$= \frac{1}{2}\,\frac{\Gamma\left(\frac{3}{2}\right)\cdot\Gamma(2)}{\Gamma(7/2)}$$

$\Gamma(n) = (n-1)\Gamma(n-1)$

$\Gamma(m+n) = (m+n-1)\Gamma(m+n-1)$

$$= \frac{1}{2}\,\frac{\frac{1}{2}\cancel{\Gamma\left(\frac{1}{2}\right)}\cdot 1!}{\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\cancel{\Gamma\left(\frac{1}{2}\right)}} = \frac{1\times\cancel{2}\times 2}{\cancel{2}\times 5\times 3} = \frac{2}{15}$$

Egs

① EV. $\int_0^{\pi} \sin^2\theta\,(1+\cos\theta)^3\,d\theta$

Soln: we have

$$I = \int_0^{\pi} \left(2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\right)^2 \left(2\cos^2\frac{\theta}{2}\right)^3 d\theta$$

$[\because \sin 2\alpha = 2\sin\alpha\cos\alpha \;\&\; 1+\cos 2\alpha = 2\cos^2\alpha]$

$$\Rightarrow I = \int_0^{\pi} 2^2 \sin^2\frac{\theta}{2}\cos^2\theta/2 \cdot 2^3 \cos^6\frac{\theta}{2}\,d\theta$$

$$= 2^5 \int_0^{\pi} \sin^2\frac{\theta}{2}\cos^8\frac{\theta}{2}\,d\theta$$

put $\frac{\theta}{2} = t$

$\Rightarrow \theta = 2t$

$\Rightarrow d\theta = 2\,dt$

θ	0	π
$t = \frac{\theta}{2}$	0	$\pi/2$

$$\therefore I = 32 \int_0^{\pi/2} \sin^2 t \cos^8 t \cdot 2\,dt$$

$$= 64 \int_0^{\pi/2} \sin^2 t \cos^8 t\,dt$$

$$= 64 \cdot \frac{1}{2} \cdot \beta\left(\frac{3}{2}, \frac{9}{2}\right)$$

$$= 32 \cdot \frac{\Gamma\left(\frac{3}{2}\right)\Gamma\left(\frac{9}{2}\right)}{\Gamma(6)}$$

$$= \frac{32}{5!} \quad \frac{1}{2}\Gamma\left(\frac{1}{2}\right) \frac{7}{2}\cdot\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\Gamma(1/2)$$

$$\Rightarrow I = \frac{7\pi}{8}$$

2. Ev. $\int_{-\pi/4}^{\pi/4} (\sin\theta + \cos\theta)^{1/3}\, d\theta$

$\left[\left(-\pi/4,\ \pi/4\right)\right.$

$\rightarrow (0,\ \pi/2)$

$\theta + \alpha,\ \alpha = \pi/4.$

$\sin\theta + \cos\theta$

$\left(\sin\theta \cdot \frac{1}{\sqrt{2}} + \cos\theta \cdot \frac{1}{\sqrt{2}}\right) \div \frac{1}{\sqrt{2}}$

$= \sqrt{2}\left[\sin\theta \frac{1}{\sqrt{2}} + \cos\theta \frac{1}{\sqrt{2}}\right]$

$= \sqrt{2}\left[\sin\theta \sin\pi/4 + \cos\theta \cos\pi/4\right] = \cos\left(\theta - \frac{\pi}{4}\right)$

or $\sin\theta \cos\pi/4 + \cos\theta \sin\pi/4 = \sin(\theta + \pi/4)$

Soln:

$$I = \int_{-\pi/4}^{\pi/4} (\sin\theta + \cos\theta)^{1/3}\, d\theta$$

$$= \int_{-\pi/4}^{\pi/4} (\sqrt{2})^{1/3} \left[\sin\theta \cdot \frac{1}{\sqrt{2}} + \cos\theta \cdot \frac{1}{\sqrt{2}}\right]^{1/3} d\theta.$$

$$= 2^{\frac{1}{6}} \int_{-\pi/4}^{\pi/4} \left[\sin\theta \cos\frac{\pi}{4} + \cos\theta \sin\pi/4\right]^{1/3} d\theta$$

$$= 2^{1/6} \int_{-\pi/4}^{\pi/4} \left[\sin(\theta + \pi/4)\right]^{1/3} d\theta$$

put $\theta + \pi/4 = t$

$\Rightarrow \theta = t - \pi/4$

$\Rightarrow d\theta = dt$

θ	$-\pi/4$	$\pi/4$
$t = \theta + \pi/4$	0	$\pi/2$

$$\therefore I = 2^{1/6} \int_0^{\pi/2} (\sin t)^{1/3}\, dt$$

$$= 2^{1/6} \cdot \int_0^{\pi/2} \sin^{1/3} t \cdot \cos^0 t \, dt$$

$$= 2^{1/6} \cdot \frac{1}{2} \beta\left(\frac{\frac{1}{3}+1}{2}, \frac{0+1}{2}\right)$$

$$= 2^{-5/6} \cdot \beta\left(\frac{4}{6}, \frac{1}{2}\right)$$

$$\Rightarrow I = \underline{2^{-5/6} \cdot \beta\left(\frac{2}{3}, \frac{1}{2}\right)}$$